

RFP



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Request For Proposal

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Abstract

The Balmy Beach Community Day Care is a collaborative community of staff, children, and parents committed to providing quality early childhood education. Balmy Beach day care is a non-profit, school-based child care center serving children aged 4 -12 across seven classrooms and operating within shared school spaces. Staff are the key primary stakeholders, managing split shifts, supervising rolling arrivals and departures, while maintaining structured activities and outdoor play.

Part of providing a quality day care experience is taking children on field trips. However, field trips are often a source of logistical nightmares for staff. For Balmy Beach Community Day Care, part of the onboarding process for these field trips is adding an identification wristband to every child attending. This process adds additional workload and pressure on already strained staff members.

This RFP identifies an opportunity to redesign the wristband application process, reducing the time taken as well as the cognitive load on staff so that they can focus more fully on supervising children and maintaining safety.

We encourage responding engineers to not just improve a specific portion of the process but also help improve the experience of children and staff alike.

Table of Contents

Abstract.....	2
Table of Contents.....	3
1. Introduction: Purpose of this RFP.....	3
2. Team Positioning in Relation to Balmy Beach Day Care.....	3
3. Overview of Balmy Beach Day Care.....	4
3.1 Day Care Classrooms & Layout.....	5
4. Typical Day: Kindergarten & School-Age Staff.....	7
4.1 Before School Program Daily Routine (7:30 am - 9:00 am).....	8
4.2 After School Program Daily Routine (3:15 pm - 6:00 pm).....	8
5. Staff Members' Morning Routine Before Leaving for a Field Trip.....	9
6. Current Wristband Application Process and Problem Description.....	11
6.1 Wristband Description and Usage for Balmy Beach Day Care.....	11
6.2 Wristband Application Process.....	12
6.3 Wristband Application Opportunity Description.....	12
7. Stakeholder Analysis within the Day Care.....	12
7.1. Primary Stakeholders: Staff and Children.....	13
7.2. Secondary Stakeholders: Parents, Staff and Teams.....	13
7.3 Stakeholder Relationships.....	14
8. Wristband Application Opportunity.....	14
8.1. Research on Supervision and Cognitive Load during Day Care Departure.....	14
8.2. Framing: Design Bracelet application during group departure.....	15
8.3. Constraints and Design Space within the Wristband Application Workflow.....	15
9. Needs, Goals, and Objectives.....	16
Goal 1.....	17
Goal 2.....	18
Goal 3.....	19
Goal 4.....	20
10. Reference Designs.....	22
10.1. Cool Maker Automatic Toy Bracelet Maker.....	22
11. Conclusion.....	23
12. Bibliography.....	23
13. Appendix.....	26
A) Interview Responses.....	26
B) Images of Balmy Beach Community day care.....	37
C) Justification of Goals.....	38
14. Source Extracts.....	39



1. Introduction: Purpose of this RFP

The purpose of this Request for Proposal is to define an engineering design opportunity to improve the operational efficiency, safety and staff workload management regarding field trip wristband applications at Balmy Beach Day Care. Balmy Beach Day Care is a non-profit child care centre that operates within Balmy Beach Public School in southeast Toronto, serving multiple age groups while emphasizing community inclusion and relationship building. The stakeholders include staff, children and families who rely on the day care's routines and supervision. This RFP focuses on a feasible, well-scoped opportunity for first-year Engineering Science students to potentially aid this community.

2. Team Positioning in Relation to Balmy Beach Day Care

We are a team of Engineering Science students engaging with Balmy Beach Day Care as part of our academic program. Our goal is to learn how engineering and design insight can support real communities in practical and meaningful ways. One member of our team previously attended this day care and introduced them as potential collaborators. Our role is to listen, observe, and learn from the staff, families, and the environment in order to better understand their daily operations, constraints, and values.

We chose to learn about Balmy Beach Day Care because of its strong commitment to children, their families, and the local community. We are especially drawn to the day care's focus on building strong relationships and its role as a shared and collaborative space within the school environment, supporting both education and social development. As an early learning setting, it represents a child's first steps towards learning and social development. We value the opportunity to support that process in a respectful and appropriate way.

As a team, we acknowledge that our position as engineering students introduces assumptions and potential biases. In particular, we may default towards technically driven or systems-based solutions, which could overlook social or organizational aspects of the day care

environment. Additionally, having a prior personal connection to the day care may create positive bias that could limit critical evaluation of existing challenges. To mitigate these biases, we ground our ideas and design decisions in observation and dialogue with stakeholders, prioritizing the lived experiences of children, staff, and families over purely technical considerations.

Several team members bring prior experience or personal motivation related to working with children or engaging with community-based organizations. As a team, we share a genuine interest in supporting children and contributing positively to their school day. These values shaped our interpretation of the RFP by guiding us towards solutions that are context appropriate, minimally disruptive and supportive of existing dynamics. Our engineering priorities, therefore, focus on feasibility, inclusivity and meaningful improvement to daily experiences.

3. Overview of Balmy Beach Day Care



Figure 1: Balmy Beach Community Elementary School and Day Care Centre.

Balmy Beach Community Day Care is a non-profit, unionized school-based child care centre. Located at 14 Pine Avenue in the Beaches, Toronto, Balmy Beach day care's mission is "to provide a safe, positive, encouraging, stimulating and bias-free learning and care environment" [12]. Balmy Beach Community Day Care operates within and shares classrooms with Balmy Beach Community Elementary School. The day care has a total of 15 staff members and a total licensed capacity of 166 children. It operates full year, which includes during the school week, PA days, school breaks and summer [12]. The day care states on their official website that they divide the children into three groups:

1. Preschool

- 2.4 to 4 years of age.
- Full-day care (7:30 am - 6:00 pm).
- Capacity of 24 children.
- 1:8 staff to child ratio.

2. Kindergarten

- 4 to 6 years of age.
- Before & After School Program (7:30 am-8:45 am & 3:15 pm-6:00 pm).
- Full-day care on PA days, school breaks, and summer.
- 30 children in total.
- 1:13 staff to child ratio.

3. School Age

- 6 to 12 years of age (Grade 1-6).
- Before & After School Program (7:30 am 8:45 am & 3:15 pm-6:00 pm) .
- Full-day care on PA days, school breaks, and summer.
- 75 children in total.
- 1:13 staff to child ratio.

3.1 Day Care Classrooms & Layout

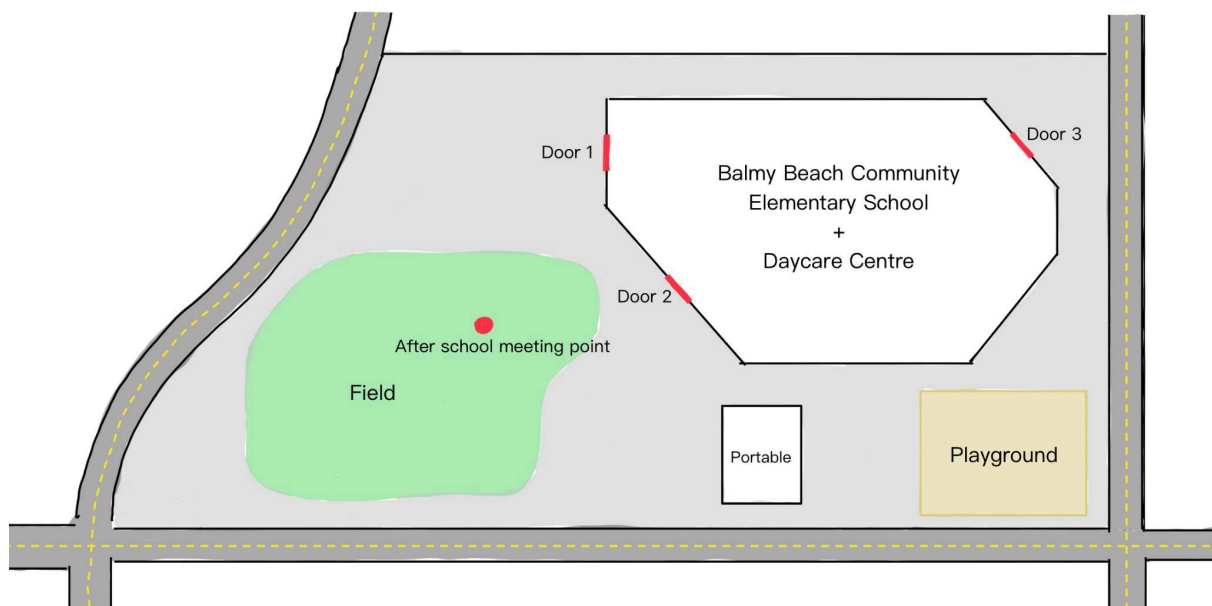
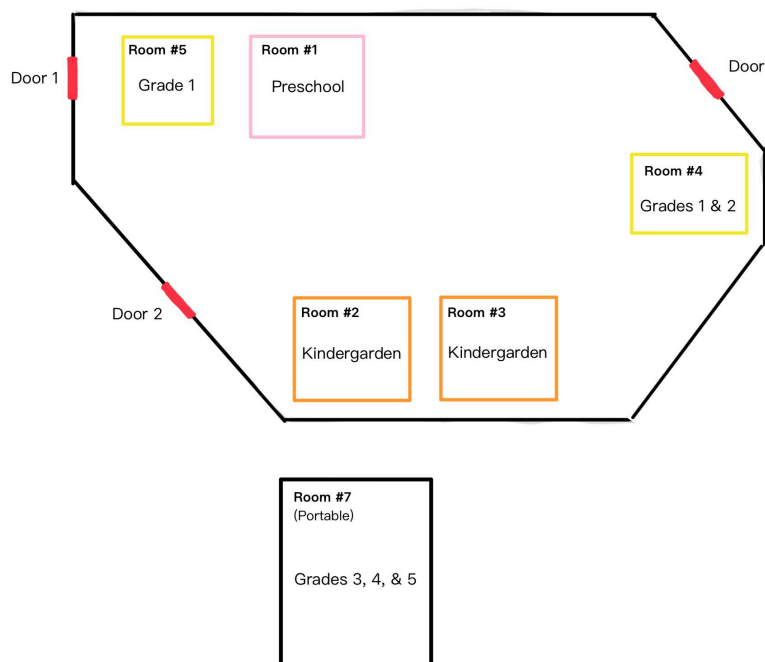


Figure 2: Layout of Balmy Beach Community Elementary School & Day Care Centre.

First Floor



Second Floor

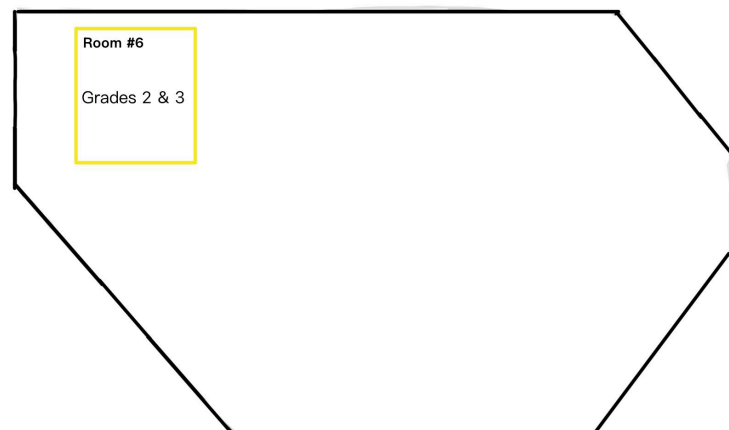


Figure 3: First Floor of Balmy Beach Day Care.

Figure 4: Second Floor of Balmy Beach Day Care.

Balmy Beach Day Care operates seven total classrooms within the elementary school. With the exception of the preschool classroom, six of the seven day care classrooms are shared with the elementary school. These shared rooms are used by school teachers during the school day from 8:42 am - 3:15 pm, and are used by the day care before school from 7:30 am - 8:42 am and after school from 3:15 pm - 6:00 pm. Based on the interview of the daycare supervisor and assistant supervisor, Christine Roberts and Patti Rychman, respectively, found in Appendix A, the seven day care classrooms are divided as follows:

- **1 Preschool classroom (Room #1):** The classroom is used full-day by the day care (7:30 am - 6:00 pm).
 - 2 staff members on-call at all times.
 - Located on the first floor, as seen in Figure 3. An image of the room is shown in Figure 5.



Figure 5: Preschool classroom

- **2 Kindergarten classrooms (Rooms #2 & #3)**

- Used by day care before/after school.
- 2 staff members per classroom.
- Both rooms have 15 students.
- Both rooms are located on the first floor, as seen in Figure 3.

- **4 School-Age classrooms (Rooms #4 to #7)**

- Used by day care before/after school.
- 1-2 staff members per classroom.
- Room #4 is Grade 1 & 2 split (15 students per grade), located on the first floor.
- Room #5 is for all Grade 1s (15 students), located on the first floor.
- Room #6 has 5 Grade 2's, 10 Grade 3's (15 students total), located on the second floor.
- Room #7 has 5 Grade 3s, the majority Grade 4 & 5, located in the portable (see Figure 6).



Figure 6: Portable (Room #7)

4. Typical Day: Kindergarten & School-Age Staff

A staff member who works with kindergarten and school-aged children works a split shift, arriving 5 minutes before their shifts begin. Their shift for the Before School Program is from 7:30 am - 9:00 am, and their shift for the After School Program is from 2:00 pm - 6:00 pm, totalling 5.5 hours of work a day. See Appendix A for the interview transcript regarding all information in this section.

The day care operates on a rolling arrival and departure system, meaning that children may be dropped off or picked up at varying times during the program's hours of operation. As a result,

staff must continually update attendance records, maintain supervision ratios, and manage transitions as children enter and leave the classrooms throughout the day. A typical day for kindergarten and school-age staff goes as follows:

4.1 Before School Program Daily Routine (7:30 am - 9:00 am)

1. Staff arrive at 7:25 am (5 minutes before their shift starts).
2. Staff read the daily written record (logbook).
 - Includes:
 - i. Messages about children
 - ii. Any unusual circumstances
 - iii. Relevant updates for the day
3. Staff are assigned to a specific room for the day.
4. Staff take attendance on a rolling basis as children are dropped off.
5. Staff supervise the children in their respective classrooms.
 - Mornings are primarily free play for the children.
 - There is minimal preparation for staff as there are no set activities due to the short timeframe.
6. Toys must be cleaned up by 8:30 am.
7. Staff help children prepare for school, which starts at 8:42 am.
 - Children are signed out properly.
 - Staff ensure children are dressed for the outdoors (e.g. children are wearing snowsuits)
 - They bring the children outside so that the children can line up with their respective school classes before the school bell rings.

4.2 After School Program Daily Routine (3:15 pm - 6:00 pm)

1. The staff's second shift begins at 2:00 pm.
2. Before school ends at 3:15 pm:
 - Staff must plan and prepare activities for the children.
 - Staff are assigned to three different school exit doors (see Figure 2).
3. Staff collect the children and take attendance at each door.
4. Once all children are accounted for:
 - Groups meet in the center of the schoolyard (see Figure 2)
 - Children are transferred to their correct day care teacher(s).
5. Staff bring the children to their assigned classrooms.
6. They require/assist the children in washing their hands.
7. Staff provide optional snacks from their rotating snack program.
8. Staff facilitate structured activities.
 - Includes:
 - i. Art

- ii. Puzzles
 - iii. Games
- 9. Children are taken outdoors:
 - Minimum 30 minutes.
 - All classes are held outside simultaneously if under 90 children.
 - Program-based activities are also done outdoors (soccer, skipping, etc.).
- 10. Staff continue facilitating structured programming until 6:00 pm.
- 11. All equipment and materials are cleaned up by the end of their shift at 6:00 p.m.

5. Staff Members' Morning Routine Before Leaving for a Field Trip

The morning before a field trip requires extensive preparation and a substantial number of tasks. There are 6-8 staff members who attend field trips. As stated in the interview in Appendix A, the morning goes as follows:

1. Before Field Trip / Early Morning Preparation

- Staff prepare as much as possible ahead of time.
- Packing materials, organizing forms, and planning roles.
- Trip description forms are partially pre-filled (child physical descriptors). Only children's clothing must be recorded on the day of.
- This preparation helps, but it cannot eliminate the morning workload because the trip description forms, wristband application, and attendance must be done live.

2. 7:30 am - 8:30 am: Arrival Processing

- Children begin arriving on a rolling basis.
- Division of tasks between staff is as follows:
 - 1 staff member takes attendance as children arrive.
 - 1 staff member must prepare emergency cards, medications, busy confirmation, and make an attendance copy (one stays, one travels). This mandatory safety work must be completed before leaving.
 - 1 staff member is assigned to complete the trip description forms. As mentioned, the form is pre-populated with physical descriptors, but staff must write what each child is wearing on the day of. If there isn't enough time, staff sometimes take the form onto the bus and finish it during travel.
 - 1 staff member must confirm permission forms and check lunches and belongings. Missing forms must be completed before departure.

- 1 staff member is assigned to apply wristbands to each child. The wristband includes the daycare's name and phone number as a safety precaution in case a child gets lost. This is a hands-on task that must be done for each child while managing movement, distractions, and late arrivals among children. This staff member is also responsible for washroom visits and ensuring that each child is wearing their trip shirt.
- Remaining staff supervise children, provide snacks, and manage behaviour. This limits the number of staff available for preparation tasks.
- Since children do not arrive all at once, unpredictable arrivals make it impossible to process all children quickly.
- Since each child must be processed individually, this further slows preparation.
- Once the room that children are gathered in reaches 30 children, a new room is opened. This creates movement between multiple rooms, and supervision across the rooms adds complexity.

3. 8:30 am: Official Drop-Off Deadline

- Families are told to arrive by 8:30 am.
- However, some families arrive right at or slightly after the deadline due to various constraints within their personal lives.
- Late arrivals restart preparation steps and extend the timeline.

4. 8:45 am: Planned Departure Time

- All staff and children are expected to board the bus at 8:45 am.
- This assumes that preparation is completed within 15 minutes after the drop-off deadline.
- In practice, this is rarely achievable due to ongoing arrivals and preparation tasks.

5. 8:45 am - 9:30 am: Continued Preparation

- Due to late arrivals, staff may still be applying wristbands, documenting clothing, or dealing with last-minute issues.
- Staff are still processing children after the scheduled departure time.
- Once all children have arrived, staff confirm attendance copies, emergency cards, medications, and child readiness before leaving. As safety checks cannot be skipped, departure waits until this is complete.

6. 9:30 am: A Typical Realistic Departure Time

- All staff and children board the bus and leave.

- Departure is often significantly later than scheduled as a result of cumulative delays: rolling arrivals, paperwork, and the wristband bottleneck.

6. Current Wristband Application Process and Problem Description

6.1 Wristband Description and Usage for Balmy Beach Day Care

As stated in Section 5, Balmy Beach Day Care makes a conscious choice to utilize wristbands for clear identification purposes. Each wristband has the daycare's name and phone number, allowing for a straightforward identification process if a child gets lost and is found by a non-daycare member. Wristbands are used because they provide continuous, on-body identification that stays with the person at all times and can be checked instantly without searching belongings or clothing, which is why they are standard for safety identification in healthcare and other high-risk settings [13]. The wristband that gets applied to each child is an adjustable snap-closure ID wristband. An image of a child wearing the Balmy Beach Day Care wristband is shown in Figure 7, and the dimensions are shown in Figure 8. Furthermore, the wristband is laminated to make it waterproof so that it can stay on during water activities (see Appendix A).



Figure 7: A child wearing the Balmy Beach Day Care wristband.

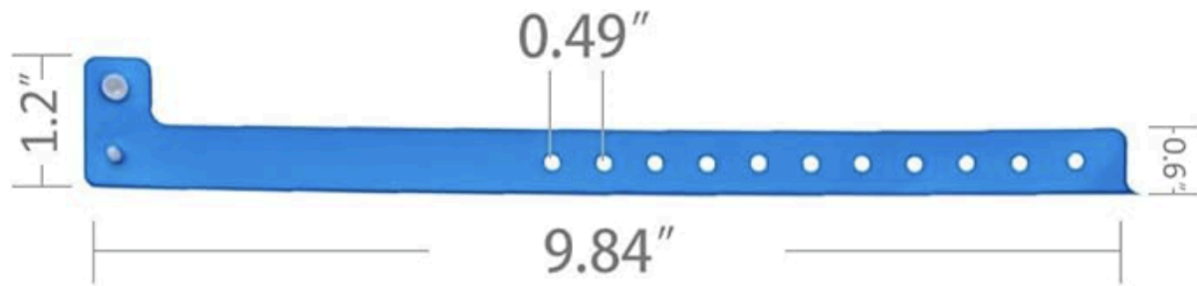


Figure 8: Dimensions of adjustable snap-closure ID wristband [14].

6.2 Wristband Application Process

The application of wristbands is currently a manual, per-child process that occurs while children are active, mobile and often excited (see Appendix A). One staff member must:

- Physically manage and gather individual children.
- Wrap the wristband around the child's wrist.
- Fast button and cut excess material using scissors.
- Confirm that each child has received a wristband.
- Simultaneously, continue supervising the remaining group of children.

6.3 Wristband Application Opportunity Description

In practice, this creates challenges reflected in the lived experience of staff:

- Time-intensive: For large groups, this can take a significant portion of a limited prep time.
- Inefficient: Manual response application requires staff to stop other tasks like supervising children, conducting headcounts, filling out forms, to focus on one at a time, a physical application.
- Safety critical: while wristbands are intended to enhance safety, the current manual process reduces staff availability for supervision. Moreover, it requires keeping scissors and other sharp objects near rowdy children.

When scaled to large groups, 90+ children, this process becomes time-intensive and cognitively demanding, requiring staff to divide attention between hands-on wristband application and active supervision.

7. Stakeholder Analysis within the Day Care

This section identifies and describes the primary and secondary stakeholders involved in or affected by the field trip departure process within the daycare setting.

7.1. Primary Stakeholders: Staff and Children

Staff are primary stakeholders because they carry the operational and supervisory responsibility for field trips. The burden of preparing kids to leave the classroom, having everything they need for the trip, and transportation falls on their shoulders. They are directly accountable for child safety and for the successful execution of the trip itself.

Staff act as both the planners and the supervisors, absorbing the majority of the logistical and emotional labour. As a result, decisions about whether a trip is feasible are often driven by workload and risk rather than educational values alone.

For staff, the most important factor in any proposed solution is workload reduction. Their core needs include:

- Maintaining required adult-to-child ratios before leaving and during the trip.
- Managing field trip preparation within limited staffing availability.
- Ensure safety in an uncontrolled environment.

These conditions increase cognitive load compared to regular in-classroom supervision. Minor errors, such as missing forms, miscount or unclear instructions, carry significant liability.

Any proposed solution must reduce planning time, increase reliability and support supervision rather than add new steps or tools. Designs that ignore staff workload, liability concerns, or time constraints are unlikely to be adopted.

Children are also key stakeholders because the identification wristbands are applied to them, and they affect their comfort, safety and experience. Although they have limited decision-making power, their needs and perspectives strongly influence how the process unfolds and must be carefully considered. Children's influence is primarily expressed through behaviour and engagement rather than formal input. As a result, their needs are often inferred by others rather than explicitly stated.

Design solutions must support predictability, clarity and safety from the child's perspective; unfamiliar systems can increase overstimulation and emotional load. Systems that reduce confusion while maintaining engagement increase enjoyment and result in a positive experience for children.

7.2. Secondary Stakeholders: Parents, Staff and Teams

The secondary stakeholders are the day care administrators and regulatory bodies, which include the owners, directors, and childcare licensing authorities. Administrators are affected through cost, implementation, maintenance, and operational feasibility, while regulatory bodies indirectly influence the design through safety, accessibility, and childcare compliance requirements.

Parents trust the day care with their child's care while remaining external to daily operations. Their primary role is providing consent and information and ongoing communication, which influences how staff plan and manage activities.

The receiving team consists of external designers responding to the RFP, their role is to interpret the framed opportunity and propose feasible solutions within the stated constraints. The receiving teams are incentivized to propose innovative solutions that demonstrate design capabilities. Other important secondary stakeholders include the authoring team and our Praxis II team. The team continues to develop our understanding of this unique community. The team frames the problem, defines its constraints and shapes what types of solutions are considered valid. While not responsible for executing the trips, the team's position directly influences the scope and direction of the design opportunity.

Furthermore, the Praxis II teaching team is an important secondary stakeholder, as they are responsible for ensuring that the communities and opportunities identified in our RFP are legitimate.

7.3 Stakeholder Relationships

Within the day care, stakeholders are connected through shared responsibility for children's care, safety and development. Staff interact directly with children on a daily basis while coordinating with parents. Administrators provide oversight about setting policies, allocating resources and ensuring regulatory compliance, which shapes how staff carry out their roles. These relationships are interdependent, and decisions made at one level affect the capacity and experiences of others throughout the day care.

8. Wristband Application Opportunity

8.1. Research on Supervision and Cognitive Load during Day Care Departure

Childcare and elementary education settings operate under strict safety and supervision requirements, particularly during transitions such as arrival, departure, and off-site travel. Licensing guidelines require continuous supervision (staff-to-child ratios approximately 1:13) and active risk management, all of which intensifies for staff during field trip preparation. [\[1\]](#)

At the same time, research shows that staff are already working under significant workload constraints. A study by the University of Sydney found that staff spend less than 30% of their workday in uninterrupted interaction with children, with the majority of time fragmented by administrative duties and procedural tasks. This high degree of task switching contributes to cumulative cognitive load and increases the likelihood of oversight in processes that require precision and consistency. [\[2\]](#)

Studies on manual, repetitive and multi-step tasks demonstrate that operational efficiency decreases as cognitive load increases, particularly when tasks must be completed under time

pressure and while maintaining situational awareness. [3] Exactly like the situation staff at Balmy Beach Day Care find themselves in, processes that require repeated attention to detail, even when individually simple, become more error-prone and time-consuming.

Research across educational settings also suggests that technology-supported system-level interventions can meaningfully reduce workload without removing professional judgement. Studies on automated attendance and tracking systems show that they can reduce manual handling, lower error rates and improve consistency while still maintaining safety standards. [4] Similarly, field-trip planning and support tools have been shown to reduce teacher anxiety, an organizational burden, by streamlining logistics. [5]

Altogether, this research indicates that staff responsibility, cognitive load, and time pressure during transitions are a systemic issue. Engineering intervention that reduces manual handling and sequential one-to-one actions can both improve efficiency and safety.

8.2. Framing: Design Wristband Application during Group Departure.

There is an opportunity to redesign the wristband application portion of the field trip departure process so that it is:

- More time efficient- reducing the total duration of the wristband application for large groups.
- Safer, by minimizing the need for staff to divert attention from larger groups and by eliminating the use of sharp objects near kids.
- Less cognitively and physically demanding for staff during high-pressure transition periods.
- More consistent and supporting a standardized process that reduces the need for release checking while ensuring that all children receive wristbands.

The opportunity is not to replace the wristbands' informational function, but to improve how and when it is applied within the broader departure workflow timeline from the student's arrival at school to boarding transportation.

This framing translates the lived experience of staff members, managing multiple responsibilities simultaneously under strict timing constraints, into a design space that prioritizes workload reduction and efficiency of the wristband application task.

8.3. Constraints and Design Space within the Wristband Application Workflow

The framed opportunity seeks solutions that acknowledge the realities of the childcare environment where there are continuous supervision obligations, dynamic movement and limited staff availability.

This opportunity allows for a wide spectrum of engineering approaches:

- For system-based tools that reduce manual repetition.
- Concepts that can be evaluated through prototyping.
- From low-tech, high-reliability solutions to more innovative technological approaches.

Staff-to-child ratios and licensing-mandated supervision requirements are fixed and cannot be changed. The wristbands should maintain the original purpose; all wristbands must have the daycare's name and phone number. Beyond this, the concept of the wristband can not be altered, as specified by the community.

This framing is appropriate for an exploration in Praxis II. It enables designers to propose both conventional improvements and unique interventions, all grounded in the lived operational realities of a child care community.

9. Needs, Goals, and Objectives

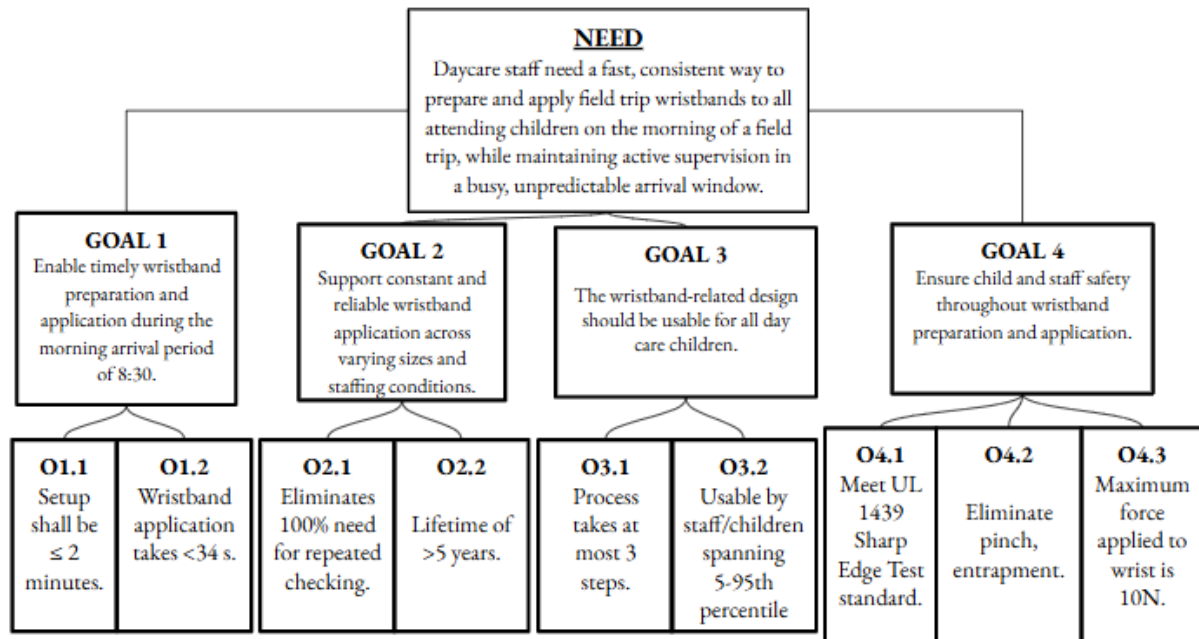


Figure 9: Flowchart Summary of Needs, Goals, and Objectives

Need:

Day care staff need a fast, consistent way to prepare and apply field trip wristbands to all attending children on the morning of a field trip, while maintaining active supervision in a busy, unpredictable arrival window.

Justification of goals can be found in [Appendix C](#).

Table 1: Goal 1 and associated objectives for designing a wristband application system.

	Description	Metric	Target	Justification
Goal 1	The design should enable efficient wristband preparation and application during the morning arrival period.			
Efficiency				

Objective 1.1 (O1.1)	Design setup shall be ≤ 1 minute of active staff time per field trip (for up to 90 wristbands)	Minutes (mins)	≤ 1	Setup time is limited to ≤ 1 minute because this limit classifies the device as a One Touch Exchange of Die (OTED) according to Kaizen Institute, which means that this setup time prevents adding a meaningful new task to staff workload [15]. Since the setup cannot meaningfully disrupt workflow, staff will have more time to focus on other important tasks like attendance, trip paperwork, washroom visits, and supervision.
Objective 1.2 (O1.2)	The design shall allow wristband application to take < 34 seconds.	Seconds (s)	< 34	The wristband application process was selected to take less than 34 seconds based on a Methods-Time Measurement (MTM) analysis, which predicts that applying an adjustable snap-closure ID wristband to a cooperative individual under normal conditions requires approximately 34 seconds. This value represents an ideal baseline time assuming minimal resistance or interruption [16]. However, interviews with one current and one former day care staff member indicated that the application can take up to one minute when a child refuses to remain still (see Appendix A). This suggests that real-world application time is often significantly longer than the MTM baseline due to behavioural variability. Therefore, an effective design should aim to reduce application time below the baseline cooperative time of 34 seconds to minimize delays and reduce the impact of child movement on the overall process.

Table 2: Goal 2 and associated objectives for designing a wristband application system.

	Description	Metric	Target	Justification
Goal 2 Reliability	Support constant and reliable wristband application across varying sizes and staffing conditions.			
Objective 2.1 (O2.1)	Enable a repeatable process that eliminates 100% of the need for staff to manually verify that each child is wearing a wristband.	Percentage (%)	100	During preparation, staff must maintain continuous supervision while simultaneously managing other tasks. Repeated verification that each child is wearing a wristband introduces additional cognitive load and task switching on staff, increasing the likelihood of oversight during high-pressure transitions. (see Appendix A for interview transcript). Therefore, a process that is inherently self-verifying reduces the need for manual rechecking and supports consistent execution across varying group sizes and staffing conditions.
Objective 2.2 (O2.2)	Ensure that the wristband application method or design maintains safe and reliable operation for > 5 years.	Years	>5	School boards and childcare facilities plan replacements for durable assets on a five to seven-year cycle. The Toronto District Board's Equipment Replacement Plan treats programme equipment with a five-year life expectancy for budgeting purposes. [6] Therefore, a five-year service life is consistent with equipment expectations in licensed childcare and educational environments.

Table 3: Goals 3 and associated objectives for designing a wristband application system.

	Description	Metric	Target	Justification
Goal 3 Ergonomics	The wristband-related design should be usable for all day care children.			
Objective 3.1 (O3.1)	The wristband application process takes at most 3 steps per child for day care children or staff to perform.	Number of steps	≤ 3 steps	The wristband process should be executable by at most 3 steps, given the preschool cognitive limits during task execution. The youngest person who could be using the wristband mechanism would be 4 years old, and by this age and older, they should be able to follow instructions with 2-3 steps. Anyone using the mechanism above the age of 4 is expected to have a higher cognitive limit than a 4 year old [7].
Objective 3.2 (O3.2)	The design must be usable by children and staff spanning the 5th-95th percentile hand size.	Percentile range	5th - 95th	The design must accommodate users between the 5th and 95th percentile hand sizes because this range represents the central 90% of the population and is the standard anthropometric design range used in ergonomics (ISO 7250-3:2015) [18]. Designing for the full 0–100th percentile range is typically impractical and may negatively affect usability, precision, and comfort for most users due to the extreme variability in human body dimensions. Therefore, accommodating the 5th–95th percentile provides an appropriate balance between inclusivity, functionality, and engineering feasibility.

Table 4: Goal 4 and associated objectives for designing a wristband application system.

	Description	Metric	Target	Justification
Goal 4 Safety	Ensure child and staff safety throughout wristband preparation and application.			
Objective 4.1 (O4.1)	The design shall meet the UL 1439 Sharp Edge Test standard.	The number of layers of UL sensing tape that get cut through when tested with a calibrated UL Sharp Edge Tester, applying 6.67 N of force over a 50 mm stroke	≤1 (pass test)	The UL 1439 Sharp Edge Test was selected because it provides a standardized, repeatable and Canadian industry-recognized method for evaluating the safety of product edges. If a maximum of one layer of the UL sensing tape is cut, this confirms that all contact surfaces are sufficiently smooth and rounded to avoid cuts or abrasions during normal handling [17]. This requirement is especially critical in a day care environment, where frequent handling and unpredictable interactions increase the likelihood of contact with product edges, making the prevention of cuts and abrasions essential for protecting both children and staff.
Objective 4.2 (O4.2)	Eliminate exposure to pinch and entrapment hazards during wristband preparation and application.	Number of accessible pinch or entrapment points	0	The Canadian Centre for Occupational Health and Safety identifies moving parts or machinery as a category for physical hazards, which includes pinching, entanglement and trapping hazards. That can cause injury if a body part contacts or is caught between them. [11] In operation, there must be no areas present on the mechanism for children to stick body parts (for example, fingers) and get hurt/pinched. (Similar to getting fingers stuck in a door).

Objective 4.3 (O4.3)	The maximum force applied to the wrist during application should be less than 10 N.	Force (N)	<10N	Utilizing force limits from a different context, accessibility and human-factors standards specify that hardware intended for one-handed operation without tight grasping or twisting should require no more than 22 N (or 5 lb) of force [10]. Cross-referencing this value, it is clear these limits are designed to ensure a safe and comfortable interaction, especially for users with reduced strength or dexterity. In a childcare context, the 22 N value provides a safe upper bound for the wristband device components. Hence, the maximum force applied to a child's wrist through any wristband mechanism must be < 10 N, clearly below the 22 N guideline value. This ensures a substantial safety margin while simultaneously reducing the risk of discomfort or injury among children during the use of the wristband mechanism.[10]

10. Reference Designs

10.1. *Cool Maker* Automatic Toy Bracelet Maker

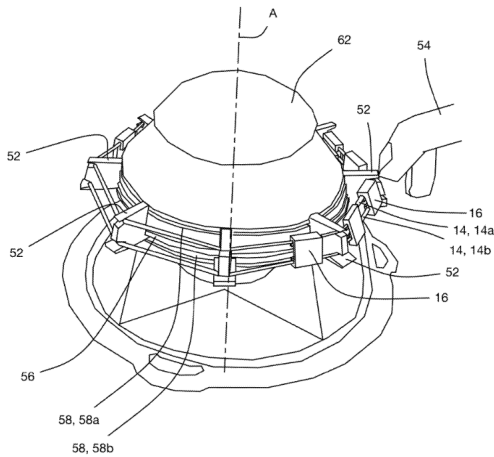


Figure 10: Diagram from Cool Maker's Patent depicting the string-bracelet making mechanism. As seen, a string runs along a circular path while a hand pulls the string out to tighten it [8].



Figure 11: YouTube video demonstrating the Cool Maker, showing the application mechanism for a bracelet-making toy [2]

Previously, a company called *Cool Maker* had made a toy bracelet maker that assists users in creating a friendship bracelet and applying it to their own wrist. While this does relate to the design we are looking for, it is built for a completely different purpose and has many shortcomings when it comes to what we would like to accomplish.

This design creates bracelets using strings and beads, while the bracelets we are using are laminated paper wristbands. As a result, a wristband solution would involve a completely different mechanism, in terms of its size as well.

It also requires human input to adjust for the wrist size, and the easiest process would require there to be enough beads on the bracelet, something that would not be present with the paper bracelets. This means that it would still require a significant cognitive load on staff, which we aim to eliminate.

11. Conclusion

This RFP identifies a clear opportunity to improve efficiency, safety and consistency of field trip application at Balmy Beach day care. Through a careful examination of the day care's daily operations, staff responsibilities and the challenges they have, we have framed a design problem that is both technically feasible and socially meaningful.

The proposed opportunity emphasizes reducing staff workload, minimizing cognitive and physical strain, while maintaining strict supervision standards and ensuring children's safety. By enabling a faster, more reliable and automated application process, the proposed solution has the potential to transform a high-pressure task into a seamless component of the day care's field trip workflow.

We invite design teams to approach this opportunity with creativity and practicality. Designs should respect the real-world constraints of childcare settings, prioritizing safety and enhancing efficiency.

In conclusion, this RFP aims to inspire engineering designs that not only address a specific operational challenge but also positively contribute to the experience of staff and children.

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13. Appendix

A)Interview Responses

INTERVIEW #1:

(informal, written in jot notes as per the interviewee's responses)

- On a side note, many notes fall under various/unrelated questions, given the interview was conversational.

Can you walk us through a typical day for different age groups?

- Important to work with the school rules
- Start 7:30 am
- Children arrive throughout the morning and are checked in by staff
- Frequent transitions between activities and shared spaces
- Meals are an important time (snack, lunch, another snack time)
- Outdoor play time requires constant safety monitoring
- Parents are the experts
- Forming positive relations with the community
- Everything is shared with the school
- There are rules within the setting, like when using the gym, for example
- Non-profit day care, supporting the community
- Support children from 2.5 to 12 years old
- Run events, more for the day care community, family barbecue in June, Christmas events, etc.
- Enrichment programs from each age group
- Outreach programs bring people in from the community
- Storage (a challenge for sure)
- Don't know where to put the storage, extra markers, glues, and all the things
- 154 kids right now
- Licensed by ministry of Education, agreement with city of Toronto
- To do better for the criteria, a challenge for the staff
- Making sure all the stuff is replenished and all
- People like the top have access to storage, not like oh it's on the third floor
- EXTRA
- Established in 1982
- Never lost a child
- But also, some kids have AirTags's
- 1:7 ratios for trips

- day care t-shirts when going out
- The City of Toronto requires trip description forms, including everything about the child, eye colour, what they were wearing, and always carries photos of children. It's a requirement, which is important.
- Leaves are a problem during the fall
- Stairs are slippery when rainy
- Accessibility is very difficult, as it is not an accessible building

- Which parts of the day are the most challenging for staff or the kids (transitions, meals, naps, outdoor play)?
 - FAMILIES: Parents want more of a Facebook group
 - Forming a connection between the families
 - Having everyone do something for themselves, not having to have one person do something
 - Having a disclaimer
 - Community contact list
 - They don't have the time to build it
 - Privacy concern, numbers, names, etc
 - Had families complete a form, but it is very difficult to get all parents to do it (about sharing contact info with the entire day care)
 - Something that is up to the parents
 - Already use an app that tells parents if they went to the washroom, what they ate, sharing all this info with parents.
 - Staff write up a little weekly report, which gets sent out every Friday
- Trying to keep the older kids engaged (like grade 4+)
- How can those kids be kept
- Dividing these kids from the younger ones
- In like small breaks, parents are often signing up for things like hockey camps.
 - They don't have time to host a camp together (within the day care) because students are signed up for these other camps.

- Where do delays or disruptions most often occur?
 - By people, when people are late, like parent teacher interview
 - Seasonal conditions, such as ice + wet floors in the winter and leaves building up in the fall
 - Anything that disrupts the flow of the day
 - Induced by people
 - Often out of control
- What safety risks require the most attention during the day?

- Playgrounds are large, but they are not usable during the winter because of the insane ice, a massive hazard.
- Sometimes it's like 17 days, they still can't go into the playground
- The obvious hallway problems, like snow in the hallway, but they try to figure out basic routines
- What activities are hardest to run due to space or resource limitations?
 - Field trips on a larger scale, from a worker's perspective
 - Staff on the floor, like taking 65 children to the center island, small problems like water, sunscreen, to ensure all children are accounted for
 - A lot of pressure on the workers
 - Taking children offsite

INTERVIEW #2 TRANSCRIPT:

Hi, this is Katie DeGregorio. Hi Katie, it's Patti, how are you? Oh my gosh, Patti! I'm good, how are you? I'm good, thank you. It's so nice to hear your voice, it's been so long.

It has been, how are things? Everything's good, I'm in my first year of university now. So I hear, That's so crazy. Yeah, so I actually came to the daycare with some of my team members in one of my classes to see Christine.

Oh yeah, that's so nice. How have you been? I'm good, I'm in the office now with Christine, so I'm not in the program anymore. Oh, okay.

So like Christine, remember how it used to be Marlene and Christine? Yes. Now it's me and Christine. Wow, where's Marlene now? She retired, she's long gone.

That's crazy, I feel like it was like yesterday that I was at Balmy Beach. Wow, it's so cool talking to you. Yeah.

Anyways, I was just wondering, do you know if Christine has like maybe like five minutes to chat with me so I could ask her a couple questions about the daycare? I'm sure she will, but she's off today and she's not in tomorrow till two, so your best bet would be to call her tomorrow after two and have a chat with her. Alright, I can let her know that you're going to call and then yeah, I'd call her after two. Okay, I actually, this is, I'm on like a little bit of a timeline because I have an assignment due.

Do you think that, do you have five minutes for me to ask you? Yeah, what do you need? Okay, amazing. So I was just wondering, okay, I'll give you a bit of background. So Christine was talking to me and my group about how it's pretty challenging to go on like large field trips because there's a lot of forms to be filled out and such in the morning before you leave.

Do you think that you could like walk me through exactly like what your day looks like in the morning before you leave for a field trip? Yeah, so it's like when the kids arrive, you have to sign them in attendance. You have to make sure if there's any permission form not signed or waiver form, depending on the place you're going. If that's not signed, that has to all get

signed.

You have to do a trip description form. So what that is, is I have to write down what you're wearing that day for every child that's going. Of course, we have to make sure kids have their lunch and all of that.

And then before we can leave, we have to come and take a photocopy of the attendance. So we leave one at the center, we take one with us. We have to make sure we have emergency cards for every kid that's going. Any medication for any child that's going on the trip has to be with us as well. Bus confirmation,

we have to have a copy of that with us. Wow, that's a lot of organization in the morning.

Yeah, definitely a lot. This form that you have to fill out about the description of what the kid is wearing, is that something that can be done ahead of time or does it have to be done in the morning? What's done ahead of time only on that form is the child's name. It's got their eye color, their hair color, their weight, and their height.

So that's pre-done, but there's a column on the side that you have to fill out the day of with everything they're wearing. Interesting. Interesting.

So how long does it take to fill out that form? Well, you'd think if we were sending, let's say all the school age is going and we have 60 kids, that's filling out that form for 60 children. So it's pretty time consuming. And how many people are filling out the form? Usually we assign it to one single person so that it's not overwhelming.

You know, one person's doing attendance, one person's doing that, one person's helping get trip bracelets on, one person's serving snacks or whatever the case may be. So it's usually only assigned to one person. So how many staff in total go on a typical field trip? I would say we usually have about eight that go in total.

Six would be regular staff, and then we usually bring in one to two staff that are just trip support staff. Okay, interesting. Yeah, thank you so much.

You're very welcome. Just in general, what do you think is the most challenging part of going on a field trip? Definitely just have to be organized. Like for me, I'm not in the room anymore, but I always tell the staff, do it the night before, get everything packed in a bag, try to be organized so that in the morning there's less to focus on.

And it's just hard because kids, you know, you may get from 7.30 to 8, nobody shows up, and you're like, okay, you're leaving at 9, but then at 8.30, 30 kids show up. Right. So that's challenging to get, like, everybody done when it's shown up like that.

It's not so staggered. But, yeah, if you can try to be organized, everything that you can organize ahead of time, as much as you can, but, I mean, it's just the way it goes. Yeah, so like what time do most of the kids arrive, or like what's the window, and then when do you guys leave for the field trips? It all varies.

It depends on where we're going. Like there's some trips where we leave at 8.15 so that we open at 7.30. That only gets 45 minutes for everybody to get in and all that to be done. Okay.

So those days are more challenging where there's other days where we're not leaving until, excuse me, 9 or 10. It's a little more relaxed because you have more time to fill it all out. Right. And kids, I can't say a specific because they can come in whenever they want. Yeah. There's no specific time to be dropped off.

Okay, that's really interesting. If there's a trip, we usually ask that you come 15 minutes early, but not everybody can do that or does that. Sometimes they come right at the time, so.

Yeah, so it varies. Yeah. But like 15 minutes doesn't sound like enough time to, example, like fill out the description form of like what the kid is wearing.

It's not. There are some times where we have to take it with us on the bus and do it when we're on the bus because there just wasn't enough time to do it ahead of time. Yeah.

So that is always an option. We can do that, but we try not to. We try to get it done before, but that is an option.

Okay, I get it. Like, sorry, what am I trying to articulate? Like, because I think that we're trying to, like, make our assignment about the process of the forms. Right.

Do you think, like, is that a particularly, like, can it be stressful or is it not that bad because you can, like, split it up between so many staff members? Like, there's a lot of people. It is still, like, yes, it's not as stressful, but it is, like, I want to say yes and no. Like, it is and it isn't.

There's just so much things to be done, and if you can delegate it and people can all jump on board and get it done, it's easier, but sometimes it doesn't always happen that way. Like, you remember there's trip bracelets that have to go on and there's shirts that have to go on, and, like, it's just a lot to remember and do. Right.

And then all the paperwork on top of it, it just adds another layer to, like, everything else you have to do. So then, like, when you have so many different staff members, like, doing different tasks, like, is it hard to then also look after the kids at the same time? Exactly. So that's the problem is that you can have a few people doing the paperwork, but somebody's still got to be watching the kids and playing with the kids and interacting with the kids.

So that's what makes it challenging, right? So, sorry, you said that there's six to eight, like, staff members. So there's six regular staff in school, right? So if they're all in, if everyone's on shift and they're in, there'll be six staff. Okay, so then how many of you guys would be, like, doing stuff in terms of preparing for the trip, and then how many of you are just looking over? Okay. So that's five staff left. Another one would do trip descriptions, so there's four left. One staff is usually doing washrooms or putting bracelets on, making sure trip shirts and all that is ready. And then the other staff are doing snack or, you know, helping the kids, playing with the kids, whatever. Okay. That is so helpful.

Thank you so, so much for answering all these questions. Do you, okay, actually, I do have, like, I do have another question that's a bit unrelated. I really appreciate your time. You're amazing. No worries. Okay.

I'm also aware that the fenced-in playground gets closed in. Sorry, like, the fenced-in

playground that's closed in, it sometimes is off limits because it gets too icy in the winter times. Yep.

So how often? And then it's closed in the springtime because it floods, because everything melts and there's no drainage. Oh, interesting. Yeah.

So, like, in the wintertime, how long in the wintertime is it closed for? Like, it just depends. If it's really depending on how icy it gets, because normally that's what the main thing is, is it's the ice, right? So if it's really cold and really icy, like there was, I think, two weeks out of this year so far that they haven't been able to use it. They're going out right now because it's not as icy and it's warmer, so things are starting to melt.

Yeah. And then I would say in the spring it's probably two to three weeks as well. All that, like, under the wooden climber floods and all where the drain floods because there's just no drainage when all the snow melts.

Right, that makes sense. So, like, is it specifically, like, on the structures, like the slides, like the stairs, is it that? No, no, that's fine. It's the surfacing.

Oh, okay. Yeah, it's the surfacing that gets icy. So it's not, the kids can't walk because they just slip and fall all over the place.

Okay, that makes sense. Yeah. All right.

Well, that was all of my questions. Yeah, I really appreciate your time. I'm here if you think of anything else.

Pardon? I said I'm here if you think of anything else. Thank you so, so much. No problem.

It was so fun coming back to the daycare and, like, seeing so many people talking to you. Right.

Yeah, it's super cool.

It was a full circle moment for you. It really was. Like, when I came back, I could not believe how small the daycare was because when I was younger, it looked so big because I was tiny.

Yeah. Yeah, it's such good memories. Your phone's cutting in and out on me here.

Oh, I just said it was really good memories. I can't hear you. Oh.

Okay, you can hear me now. Good. I just said that I have a lot of good memories, and I definitely remember you.

I loved you. You were, like, my favorite. I see your brother every so often, but I haven't seen you forever. I know. I don't stop by very often. I need to more.

Yes, you do. Come by and see us soon. I will.

Okay, well, if you think of anything else, you let me know, or if you need to talk to Christine, she'll be in after two tomorrow. Thank you so much, Patty. No problem.

Have a great night. You too. Bye.

Okay, bye-bye.

INTERVIEW #3:

Hello, this is Katie DeGregorio. Oh, hey, how are you? I'm good. How are you? Good.

I'm so sorry. I got your message after six the other night, and I forgot yesterday, but you didn't leave your phone number. So I thought, oh, I can go back through call display and find it, and I forgot.

Oh, don't even worry. I didn't even think to leave a phone number. That is really smart.

How are you doing? Pretty good. Awesome. You had a nice day off? Oh, I called yesterday, and Patty picked up, which was so nice.

I wasn't off. I was just working from home for a couple of hours, I think. Oh, that's nice.

I got him. I did the afternoon. Oh, no.

You know what it was? What was it? Yesterday, I opened, and I left early to drive my son to the airport because he was leaving for Europe. That's what it was. Wow.

Wait. Is he back? He's just going on a trip? Yeah. So he's third-year university.

He's just adding a couple of extra days to reading week. Nice. And, yeah, he's gone.

That's super fun. Wow. Mm-hmm.

So what were the questions that you had? Awesome. So you're okay to chat for five minutes?

Mm-hmm. Thank you so much.

Okay. I was just wondering. Okay.

So basically, we submitted our original assignment, and part of the feedback that we got was that we need to be more detailed with what we're going to be doing. So, in terms of a typical day, do you think that, from a staff member's perspective, could you walk me through what a day looks like? From whose perspective? From a staff... From the staff perspective? Yeah, staff. Okay.

So employees arrive five minutes before their shift. Okay. They make sure that they read the daily written record, which is a logbook, with anything specific.

you know, that might affect them for the day, for their class. Read daily records? They're reading daily records. Okay.

So, it's any messages that could be pertinent to the children in their class, or anything out of the ordinary, right? Okay. So that's written in there. You know, they put all their codes, all their personal belongings, it all gets locked up, and they're ready to start their shift for the scheduled time.

And most staff work as a team. They all have a schedule, kind of how the day goes, shift is.

So, you know, if the children are very young, they have set scheduled times for toileting and eating and that kind of thing.

Each classroom, regardless of the age, all provides a daily program plan. It's completed by the staff for the week, but each day has something planned pretty much for every developmental area. So those things are facilitated throughout the time that the children are in care.

So the staff prepare those materials, set those activities up, lead it, assist the child as needed, that kind of thing. So whether it be crafts or science experiments or any of those kinds of things. Again, depending on the age, it may be responsible to lead circles.

So it's kind of a learning circle. So just so I understand, what age group goes on field trips?

Primary. Okay, so for a staff member that is with, we'll say kindergarten, what time would they typically arrive at school? Yeah, the staff member.

Oh no, not on a trip, just on a typical day. I was just curious about that. It's different for everyone.

The center opens at 7.30. On a school day, most staff come at 7.30. Some work a split shift, so they're only here until 9 or 10, up to 6. But there's only three of them. Mostly almost everybody that works here works a split shift. So most people work from 7.30 to 9, leave, come back from 3 to 6. Yeah, but the average day for a split shift employee is five and a half hours.

So if they come back at 2, they leave at 9. If they leave at 10, they come back at 3, that kind of thing. Right. Okay, that makes a lot more sense.

So from 7.30 to 9, for kindergartners, let's say, they obviously have school. So parents drop off their kids early. And before school, what are the staff members doing? Mostly it's not programmed activities.

It's usually more free play because the time frame is not very big. Right. Because the kindergarten also happens in the same room as the school classroom.

The teacher, ideally, probably by 8.30, would like our stuff cleaned up. So by 8.30, our toys and things are put away. They're on a rolling cart if we live next door.

And between that 8.30 and 8.40, when school is to start, kids are usually getting into snowsuits if their particular class goes up to play first. Or they may be, if their class doesn't go up to play first, meaning at the school bell, then the teacher may do some stories, that kind of thing with them. Or they may go to the book center for the last 10 minutes of the morning.

But yeah, before school, there's not a whole lot of planning. That seems to come later when there's more time. Because it's only 7.30 to 8.42. Right.

While we have children. And then that remaining, you know, sort of 17, 18 minutes, is the staff, you know, making sure that all those children are marked out and it's reported to the teacher often who's away. You know, that kind of thing.

Okay, so remind me, the room that you showed me, like where you work, that's for preschool, correct? You got it. That's for 2.5 to 3.8 year olds. And they're here all day.

Okay. And then there's, I believe, two other classrooms like inside of the school. There's actually more than that.

So there's two kindergarten classrooms inside the school, which is a shared space that the school teacher uses at 8.45 to 3, and we use it the other times of the day. And then there's also two school-age classrooms inside the school, same component, like timeline. Really? And then on the second floor, there's another school-age and there's a portable.

Like we have seven classes. I did not know that. Wait, so.

So four are school-age, two are kindergarten, and one is preschool. Four school-age, two kindergarten, one preschool. Okay.

Yeah. So two school-age on the first floor, one school-age on the second floor, another school-age in the portable. You got it.

Okay. So then, so like each, you kind of just get assigned a room. You got it.

So all staff are assigned to a classroom. Okay. That makes so much more sense.

And does like the staff member like get like the same group of children every day? Yes. Got it.

So because there's four school-age classrooms, is it like divided up by like, okay, like this classroom for grade one to two? 30 in 104, 15 in 100, and 15 on the second floor.

Okay. Is like, is each class, like is one, is one for like grades one to two, another for like grades three to four? We try. We try.

Like we have one, one room, 100, 15 kids. It's all grade ones. Got it.

But 104, it's grade ones and twos, like split 15 of each. And then upstairs, we have a very small portion of grade twos, like five and 10 grade three. And then in the portable, we have about five grade threes and a big portion of fours and fives.

So it's really expanded since I graduated. Yeah, for sure. Wow.

That's not how I remember, but that's awesome. Like, I mean, everyone has such good things to say about the daycare. So I'm not surprised that everyone wants to send their kids there. Well, okay. This is really helpful. And then just so I have a bit better of an understanding.

So like after school, when the kids, like the kindergarten slash school age kids come back from school, like what do you, what do staff members do afterwards? The staff actually pick the kids up. So three doors basically, three different entry points. So we have staff assigned, you know, put in the line, the attendance is taken, that kind of thing.

And then once each door has all the kids accounted for, they all meet in the center of the schoolyard, and the kids are given to the proper teacher. Because unfortunately, it's like we might go to door number one, but she may have kids that come out of door number one and door number two, but she can't be in both places at once. Right.

So it's very, you know, very tight on, you know, who comes at what door, who's responsible to find them. And then once we have everybody, we join, and they go to their proper teacher into their assigned class. Okay.

And then what do you do? Everybody enters their classroom. You know, they do a little hand washing.

Snacks are provided, you know, for anyone who wants to eat.

It's like a rotating snack kind of deal. If you don't want to eat, you don't have to. You can go right to activities.

And then that carries on, you know, for a while. There's outdoor time also. In the afternoons, they go out for a minimum of a half an hour, and all the classes go out.

So, you know, if there's 90 or less, the kids are away. So they're all outside at one time. And then program plan activities are outside too.

So there'll be some playing games like soccer or, you know, skipping, whatever, right? So and

then when they're inside, they've got program plan activities as well. It's not in the afternoon. It's not free play.

Okay. All right. And then by 6 p.m., like everyone, like the staff, like clean up whatever and go home.

Okay. That makes sense. And so like on a typical day, like how many staff are working in total? And how many like in each classroom? Okay.

That is so, so helpful. And I'm also curious, like when you guys do go on field trips, I remember you saying that it can be like 60 kids going on this field trip. Where are you guys meeting in the morning? Like are all 60 kids in one room? Okay.

So, but like which classroom, like specifically, like are all 60 kids say going to the preschool?

Right. Sorry. I'm not like, I'm not explaining my question properly.

Like if, if like, like when all of the kids come to the daycare, like are they all like in one specific room while you guys are getting ready to leave? No, but because again, because the rooms are only so big, right? But basically if, if, if they're in, let's just say they're in room 104 and there's 30 children assigned to that room with two staff, you can't put two. Right. And then we open a third room.

So it's, it's based on as we're starting to get close to like the maximum. Okay. That makes sense.

Okay. Okay. And that basically answers all of my questions because, so basically what my group, we really liked what you were like when you were talking about the field trips that you guys go on that, like, like it does sound really challenging that like there's a lot of people coming in at once, like on a rolling basis.

Yeah. Like we just, you know what I mean? To try to do our best schedule. If you come in at this time, you go at 10 30 and if you, you know what I mean? Like everything is really structured to do our best to keep it in order for, for 90 kids.

Yeah. And Patty mentioned to me that like, sometimes if kids are like showing up like right before you're supposed to leave that, like you guys are like filling out forms on the bus, like for like the description forms. Yeah.

So one of the requirements from the city is that when you take children on site, so I think I might've mentioned that when we met, you know what they're wearing that day. But sometimes, you know, like if we say be in by eight 30 people come at eight 31. So, you know, 10 people come at eight 31 on schedule or we miss our turn or, you know, wherever we're going to the venue sometimes.

Yeah. Sometimes the staff are pulling out that sheet and filling in what they're wearing after we boarded the bus. Right.

You know, like if we were to stay back 20 minutes from the parents for that reason, every time, you know, it just cuts in on the kids fun. Yeah, for sure. So like to fill out that form for one person, like how, like how long would that take you? Like, you know, 14 seconds.

Like it's, it's not a, it's not a huge deal, you know, for one, you know, for one kid. But you know,

like I said, because you know, there's lots of families coming right at the deadline. Um, you know, maybe you've got things like, um, you know, Oh yeah.

Hi. You know, we're here at eight 30. Oh, hang on.

We just have, and you know what? I don't have a water bottle. Cause I'm just going to run the left and found to see if maybe they lost theirs and left it there. So you do get this kind of stuff, um, you know, and it's very challenging cause you don't want to seem unsympathetic, but you know, the other 89 families made it on time.

Yeah. You know what I mean? So it's, it's, you know, like these are things that, that, happens or, you know, we have rules like, um, on a field trip, parents cannot drop their kids.

And we say it's safety reasons.

Like each group, you know, like for instance, if you drop a kid off and you know, we didn't know they're coming, we didn't bring their emergency card with all their information. Yeah. Like, you know what I mean? They weren't put into a group.

Um, you know, so, so that's one of the things that we try to be pretty sticky on, like not to be, um, you know, not helpful, but it's just, it's 90 kids. It's a lot. It's a lot.

It's a lot that, you know, if we've got random parents showing up, you know, Oh, here's another one. Um, you know, it's just, um, so we say, no, you need to be in on time. You can't, you know, drive to wherever and drop them off.

Right. So like, if, if you say like, okay, like come to drop your kid off at eight 30, are you guys leaving at eight 45 or nine? Usually, usually I get grace. Usually I get 15 minutes.

Um, you know, so if, you know, if I eat 30, just, you know, in case there's any kind of little issy or something, you know what I mean? And we say, Oh, trips back. Yeah. That makes sense.

So, okay. But like if you're meeting at eight 30 and leaving at eight 45, that sounds like so little time to like fill out those forms, get everything ready. Well, because most families like, like maybe we leave that early.

So when do you actually, you know what I mean? Nine 30. So, you know, everyone is normally in, um, you know, like families are used to dropping off the book quarter to nine. So even in the summer or on the day, right.

So, you know, we try not to book a ton of trips that make parents come in way before they normally would. Um, so, you know, it happens, but it's not, it's not like every day. Right.

Right. So it does give staff a little bit more time. And remember, as part of that, she's prepopulated.

So they're only writing down blue shorts, red shirt. you're right. Like literally blue shorts, red shirt.

That's it. Okay. I, I understand so much more like Christine, this is, this is so, so helpful.

Like you like taking the time to explain all this to me. I really appreciate it. Now I can like be a lot more detailed.

I guess that's it. Did you end up going to ballet school yet? No, that one didn't end up working

out. So, uh, yeah, we're, we're writing about you guys.

And it's really, it's really good. Like we're, we really liked the opportunity with field trips. Like it's like a good, like, cause it's so detailed.

It's like such a specific process that there's like a lot to write about and a lot to, well, you know, another little bit is on field trips, staff can't take trips. Like that might be something that you would want to, right. So, you know, like when, when they go off the trip, you know, from eight 30 to three 30, but you know, you don't get much break or again, we do our best when they get back to get them something.

Um, but, but yeah, there's no, no break. Yeah. That makes sense.

Like, I think that like what we're going to like, kind of like hone in on is that like, there's, it's just like probably like a, there's a lot to do in that, like whatever half an hour before you leave. So like, if there's ways for us to, this is a really busy day. We might bring some stuff in a little earlier.

Like if that's like 10 to six, if the, you know, if the kids are in at eight 30, they have, we do have extra, you know, we have pretty much our full staff, um, you know, which, which is a little easier because you know, if they're all outside, one of them could be kids, you know, outfits, you know, that kind of thing. So they can, you know, kind of try to divide and conquer. Yeah.

All right. Well, yeah, I hope I, I really, really appreciate this conversation and I'm really excited to start writing about it. No problem.

And it's something, you know, if you decide you are going to go with us and you have other questions, I don't know. Awesome. Christine, thank you so much for your time.

No problem. Take care. Take care.

Have a great rest of your day. Bye. Bye.

B) Images of Balmy Beach Community day care





C) Justification of Goals

Goal1 Efficiency	The design should enable efficient wristband preparation and application during the morning arrival period. Justification: Field trips occur during one of the most demanding transition windows in the day care setting, where staff must balance supervision and multiple procedural tasks under time constraints and staffing ratios. Continuous supervision requirements (approximately 1:13 staff to child) place constant pressure on staff, significantly limiting time.
Goal 3 Ergonomics	The wristband-related design should be usable for all day care children. Justification: As discussed, wristband application occurs during a demanding transition period where staff must supervise children while completing multiple field trip tasks. Poor ergonomic design in the wristband mechanism can increase cognitive strain or physical fatigue for teachers and students. This is especially prevalent if the design requires awkward postures while interacting with active, energetic children. This is especially important to minimize staff intervention.
Goal 4 Safety	Ensure child and staff safety throughout wristband preparation and application Justification: Childcare licensing guidelines require continuous supervision and active risk management, which becomes more difficult during transition periods. Safety-focused designs must therefore reduce attention demands and physical hazards rather than add them. [1]

14. Source Extracts

Schedule 1: Requirements re Child Care Centres

	COLUMN 1	COLUMN 2	COLUMN 3	COLUMN 4	COLUMN 5
Item	Name of age category	Age range of age category	Ratio of employees to children	Maximum number of children in group	Proportion of employees that must be qualified employees
1	Infant	Younger than 18 months	3:10	10	1/3
2	Toddler	18 months or older but younger than 30 months	1:5	15	1/3
3	Preschool	30 months or older but younger than 6 years	1:8	24	2/3
4	Kindergarten	44 months or older but younger than 7 years	1:13	26	1/2
5	Primary/Junior School Age	68 months or older but younger than 13 years	1:15	30	1/2
6	Junior School Age	9 years or older but younger than 13 years	1:20	20	1/1

[1]

A national survey of early childhood educators revealed unsustainable workloads and unpaid hours are impacting the quality of care provided to young children.

Led by the University of Sydney, [the study](#) is the first in Australia to link educator workload with early childhood education quality. It found educators spend less than 30 percent of their time, or under 2.5 hours a day, in undistracted and uninterrupted time with children; more than three-quarters work an average of nine unpaid hours per week; and over two-thirds said their workload demands reduce the quality of care they can provide to children.

The researchers are now calling for a national review of work in early childhood services as a first step toward improving quality and ensuring workforce wellbeing and sustainability.

Workload pressures undermine quality of education and care

[2]

Impact of work instruction difficulty on cognitive load and operational efficiency

increasingly critical, especially under the human-centric principles of the 5th industrial revolution. Building on cognitive load theory (CLT), this study compares two instructional methods-visual-based and code-based-to determine whether cognitive overload can be reduced without compromising task outcomes in a controlled, assembly-like scenario derived from industrial tasks. We recruited 30 participants from the academic field (students and researchers), who completed assembly tasks under both visual-based and code-based instructions. Cognitive load was measured objectively by (Galvanic Skin Response, Heart Rate Variability, and hand motion acceleration) and subjectively through (NASA Task Load Index, short Dundee Stress State Questionnaire). Operational efficiency was assessed via task completion time (TCT), number of task repetitions (NTR), and assembly precision based on the standard deviation. The findings demonstrated that visual-based instructions significantly reduced cognitive load with a $p - value < 0.001$. It also showed an improvement in two of the performance metrics during the use of visual-based instructions for the TCT and NTR with $p - values < 0.001$. However, although code-based instructions increased cognitive load, they showed better assembly precision with a $p - value < 0.001$. These results suggest that while simple and direct instructions facilitate task execution and reduce cognitive loads, deep thinking approaches may still hold value for tasks requiring high precision.

[3]

The use of Radio Frequency Identification (RFID) technology is ubiquitous in a number of businesses and sectors, including retail sales, smart cities, agriculture, and transportation. Additionally, educational institutions have started using RFID to track student attendance, combining this technology with Google Sheets and the Internet of Things (IoT) to build a real-time attendance tracking system. For a thorough examination of the creation of a student attendance system, this paper includes a systematic literature evaluation of 21 major research published on IoT based attendance systems employing RFID. This RFID-based attendance system enables automation, eliminating several problems connected with the manual process, such as time wasting, proxies, and the possibility of losing the attendance sheet, in contrast to the traditional attendance system, which depends on manual signatures. By creating a system that automatically registers students' attendance by merely flashing their student cards at the RFID reader, all the aforementioned difficulties may be successfully addressed. This automated method guarantees attendance monitoring accuracy and dependability while also saving time. This paper's conclusion highlights the significant advantages of implementing an IoT-based attendance system based on RFID technology. The suggested solution provides a trustworthy, efficient, and secure alternative to manual attendance techniques, successfully addressing their shortcomings. This paper offers helpful insights for institutions looking to create a cutting-edge attendance system that increases student involvement and academic achievement by looking at guiding principles, best practices, and the successful resolution of difficulties.

[4]

An Intelligent Support System to Help Teachers Plan Field Trips

Abstract

Field trips enrich learning programs with out-of-school activities that can bring gains in students' academic content knowledge and personal growth. However, they are a source of anxiety for teachers because of the bureaucracy, pedagogy, etc., risks they imply. To address this issue, we propose FieldTripOrganizer, a field trip planner based on the mixed-initiative approach aimed at increasing teachers' autonomy and motivation in designing educational tours. The key aspects of our application are (i) the simultaneous provision of information filtering and automated scheduling support while the user designs the field trip, and (ii) the visual annotation of places and activities to show whether they can be included in the itinerary without violating its time constraints. Different from current tour planners, these functions enable the user to be in full control of the design process, delegating the system to manage difficult and burdensome tasks such as consistency checks and itinerary optimization. We evaluated FieldTripOrganizer in the use case of organizing a science field trip. In a preliminary user study involving 18 science teachers, our application turned up to be superior to a baseline tour planner in both usability and user experience. Moreover, the teachers declared that it was helpful, motivated them, and reduced their anxiety during the design of the field trips.

[5]

1. Why does the Renewal Needs Backlog keep increasing?

The Ministry of Education hires external consultants to inspect each school once every five years. That means that 20 percent of our schools are inspected every year. As part of this assessment, components that need to be repaired or replaced are identified.

Approximately 50 percent of our schools are over 60 years old and building components continue to age requiring major repairs or replacement.

[6]

learns a lot of new words by listening to you and others and also by listening to stories

- might like to tell stories and have conversations.
- understands many more words than they can say.
- understands most of what you say and might guess the words they don't know.

Around **3 years**, your child:

- uses sentences of 3-5 words or even more
- points to parts of pictures – for example, the nose of a cow – and names common objects
- can be understood by other people most of the time.

By **4 years**, your child:

- speaks in longer sentences of around 5-6 words or more
- follows instructions with **2-3** steps, as long as they're about familiar things – for example, 'Close the book, and give it to Mum'
- understands [adjectives](#) like 'long' or 'thin'
- uses 'feeling' words like 'happy' or 'sad'
- can be understood by most adults.

Exploring bodies and gender

Your child:

- probably [expresses their gender](#) through the way they talk about themselves and the clothes they choose
- can recognise other people's gender
- might be very curious about bodies – for example, they might look at their own and other children's genitals.

Curiosity about bodies, gender roles and sexual concepts is usually a typical part of [childhood sexual development and behaviour](#). But if you're concerned about a child's sexual behaviour, it's a good idea to talk with a GP, a paediatrician or another qualified health professional.

Moving



Figure 11

[9]

3.0 WORKSPACE, STORAGE, AND WORKSHOP DESIGN

3.1 Workspace and operations in non-workshop areas

3.1.1 Workspace

3.1.1.1 General Maintenance clearance. To provide adequate clearance for maintainers and to provide sufficient space to accommodate tools, test equipment, procedures, and other job aids during an in-place repair, a nominal 914 mm (36 in.) maintenance clearance should be provided around all major system components and piping of 610 mm (24 in.) diameter and larger.

3.1.1.1.1 Console floor space. Whenever feasible, free floor space of at least 1.22 m (48 in.) should be provided in front of consoles used by maintenance personnel. For equipment racks that require maintenance, free floor space should be provided in accordance with the criteria given below.

3.1.1.2. Depth of work area. Clearance from the front of the rack to the nearest facing surface or obstacle should not be less than 1.07 m (42 in.). Where wheelchair accessibility is desired, 1.22 m (48 in.) should be used as a minimum. The minimum space between rows of cabinets should be 200 mm (8 in.) greater than the depth of the deepest drawer (equipment) (See also Figure 3.1.1, Depth of work area).

3.1.1.3 Lateral workspace. The minimum lateral work space for racks having drawers or removable equipment should be as follows (measured from the drawers or equipment in the extended position):

- a. For racks having drawers or removable items weighing less than 20 kg (44 lb): 460 mm (18 in.) on one side and 100 mm (4 in.) on the other.
- b. For racks having drawers or removable items weighing over 20 kg (44 lb): 460 mm (18 in.) on each side.

3.1.1.4 Space between rows of cabinets. The minimum space between rows of cabinets should be 200 mm (8 in.) greater than the depth of the deepest drawer or cabinet.

3.1.1.5 Change in position. Workspace should allow the technician to change posture if the maintenance task being performed requires prolonged kneeling, crawling, or crouching.

3.1.1.6 Wheelchair access. When wheelchair accessibility is required, a clear floor space at least 762 mm (30 in.) by 1219 mm (48 in.) should be provided to allow either a forward or parallel approach by a person using a wheelchair. The required reach envelope should be from 381-1219 mm (15-48 in.) above the floor. Hardware for accessible storage spaces should be operable with one hand and should not require tight grasping, pinching, or twisting of the wrist. The force required to use this hardware should not exceed 22 N (5 lb). Touch latches and U-shaped pulls are acceptable.

3.1.1.7 Space adjacent to live electrical parts. When the work space is adjacent to live electrical parts with up to 150 volts that may require examination, adjustment, servicing, or maintenance, a minimum clearance of 3 feet shall be provided. This distance shall be measured from the live parts if they are exposed or from the enclosure front or opening if the live parts are enclosed. For higher voltages, the clearance shall conform to OSHA standard 29 CFR 1910, as applicable. When voltages are greater than 150 volts, that voltage should be specified.

What are the hazards of working around machinery?

[Back to top](#)

Machines are used to cut, shape, press, move, spin, or exert force. Industries that use machines include manufacturing, construction, machine shops with metalworking equipment, and woodshops with woodworking equipment. However, convenience stores, gas stations, food stores, department stores, warehouses and others have bread slicers, meat slicers, mixers, conveyors, cardboard crushing machines, and other similar devices that could be a hazard.

Hazards associated with working near or on machinery vary depending on the exact machine used but can include exposure to:

- moving parts (e.g., risk of injuries from entanglement, friction, abrasion, cutting, severing, shearing, stabbing, puncturing, impact, crushing, drawing-in or trapping, etc.)
- energy (e.g., electrical, electromagnetic, magnetic, etc.)
- heat or cold
- noise
- vibration
- radiation
- gas or liquid under pressure (e.g., injuries from injection or ejection by hydraulic systems, pneumatic systems, compressed air, paint sprayers, etc.)
- psychosocial hazards (e.g., stress, job content, work organization, cognitive factors, etc.)

[11]

MISSION STATEMENT

Balmy Beach Community Day Care Centre's mission is to provide a safe, positive, encouraging, stimulating and bias-free learning and care environment. Our goal is to fulfill the needs of the individual as well as the group.

[12]

In hospital settings, one of the core principles is to ensure **patient safety** and avoid harm. However, basic errors—like **incorrect identification**—still occur daily, even in advanced healthcare systems, leading to serious clinical, ethical, and economic consequences.

Among the most widely used patient identification methods are **wristbands or ID bracelets**. Their use responds to the need for a visible, practical, and accessible tool to ensure that each patient receives the correct treatment. These bands typically include information such as the patient's full name, date of birth, medical record number, and in some cases, a barcode or chip for digital reading.

[13]



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[14]

Optimize with OTED for single-touch changeovers

OTED (One-Touch Exchange of Die) is an advanced extension of the SMED methodology, aiming to achieve setups that require only one movement or a simple gesture, usually completed in under one minute. This concept takes the quick-change principle to its most extreme level, focusing on automation, total standardization, and technical integration.

Key features of an OTED setup include:

- Changeover is performed with a single touch—via a lever, button, or command.

[15]



In today's competitive landscape, both manufacturing and service industries are under immense

pressure to deliver higher productivity, lower costs, and faster turnaround times. With global supply chains becoming more complex and customer expectations rising, businesses cannot afford inefficiencies in their processes. This is why Methods Time Measurement has become an indispensable tool for organizations that want to gain an edge in operational excellence.

MTM is not just another productivity tool—it is a scientifically developed predetermined motion time system (PMTS) that provides a universal method for analyzing work. Unlike traditional time-and-motion studies that depend heavily on a stopwatch and subjective observation, MTM eliminates guesswork by relying on a catalog of basic human motions, each with a standard time value. These building blocks allow businesses to accurately calculate how long any task should take, regardless of who performs it.

Conducting an MTM time study offers multiple advantages. It enables engineers and managers to establish precise time standards for tasks, design optimal workflows, and uncover inefficiencies that often go unnoticed. For instance, in an assembly line, MTM can highlight unnecessary hand movements, awkward reaches, or poorly placed tools—issues that may appear minor individually but collectively lead to significant productivity losses. By addressing these inefficiencies, companies can achieve better workplace design, consistent output, and reduced operational costs.

[16]

How does testing work?

Performing a test for edge sharpness is simple. Run the tester at least 50 mm (2 inches) along the edge of the subject item and then back to its starting position without the removal of the tester from the edge. Per UL 1439, the tester and corresponding tape caps are calibrated with a force to simulate human touch. As a result, a dangerously sharp edge will cut through the two outer layers of the sensing tape, visibly exposing the lowest layer of tape (black) cap. If this base layer is not exposed then the test subject's edge is acceptable for consumer use.

What kinds of materials can be tested?

Just about any accessible edge surface can be tested. The testing procedure is designed to exert a specific amount of pressure on the item in order to provide consistent and accurate results. As a result, items with metal, glass and hard plastic edges make the best test subjects.

[17]

ISO 7250-3:2015(E)

3 Worldwide and regional design ranges

Design ranges presented in this part of ISO 7250 are directly related to ISO/TR 7250-2.

Design ranges are specified as smallest to largest 1st to 99th and 5th to 95th percentile value observed in national populations. Each percentile value in a design range comes from a single member body. Typically, product design requirements specify a smallest to largest range of values to be accommodated. However, some product standards specify medium size or midrange values, so midrange estimates are also provided here.

The worldwide ranges specified in this part of ISO 7250 are normative for product standards. However, some users of this part of 7250 may be developing products designed for a more focused, regional market. For this reason, a method is presented for creating regional design ranges from individual member body data presented in ISO/TR 7250-2. Examples of this process are shown in informative [Annex A](#). For the anthropometric data used in deriving regional and worldwide ranges, the procedure and information for acquiring these data directly follow from the process described in ISO/TR 7250-2. The process of specifying design ranges defined in this part of ISO 7250 starts from the individual member body data presented in ISO/TR 7250-2.

Similarly, for many applications, user accommodation is maximized by considering male and female design dimensions separately. For example, maximum accommodation would be achieved by designing to a range that extends from the smallest 5th percentile value, male or female, to the largest 95th percentile value, male or female. [Annex B](#) shows the results of this procedure and should be used when the largest accommodation range is desired.

[18]

Stakeholder Contact

Balmy Beach Community Day Care Contact Information

E-mail: info@balmybeachdaycare.com

Tel: 416-698-5668

Address: 14 Pine Avenue, Room 107, Toronto, ON M4E 1L6

Notes

- Scissors around kids, hazard
- Time consuming, if we automate this we can decrease 1 staff member (allowing 1 more staff member to actually watch the kids)

“Making this process more effective” → quantify this process in terms of currently how long the process takes (we can decrease the amount of time it takes)

What we need to do:

- Justify why this is a problem?
- NGO's
- Framing the opportunity

Professor Feedback Feb 11

It is important that solutions solve one single problem

- It is bad to have a whole bunch of solutions for a bunch of different problems

We want to frame this as a process design

- End to end from onboarding the child to getting information, etc
- We are choosing this

Or scope in to one of the components

- Getting the data specifically and inputting it into a database

Argue that the constraints of the problem are unchangeable

- Children will come whenever they want to
- Parents will not be able to send their kids earlier than they already are.

Can we come up with 5 different solutions with our framing?

- 5 different processes in 5 minutes
- Different combinations of different solutions from different parts of a process counts as different solutions
 - 2 solutions for each of 3 steps is $2^3 = 8$ solutions

Tab 3

Need 1: The Need for Speed

Staff often find themselves struggling to keep up with the great demand for children. During field trips, these staff take a long time to prepare these children for the day ahead. Due to laws in Toronto, the staff have to do these processes on the morning of the field trip, greatly constraining the time needed to prepare for every trip. Traditionally, putting a bracelet on a person's wrist takes a lot of time, which is why the improved design must be faster.

Goal 1:

The process should support staff in helping them finish their onboarding process for the children to go on the field trip.

Need 2: The Need for (Or Lack Of) the Brain

Staff often find themselves strained between multiple tasks and the children they need to worry about. Because it is mandatory to keep a 1:13 ratio of teachers to students, a teacher must not be completely preoccupied with a single job to pay more attention to the kids at hand.

Goal 2:

The second goal is to reduce the cognitive load on staff during the onboarding process.

Objective 2.1:

The

Goal 1: enable timely wristband preparation and application during morning arrival period of 8:30

- **O1.1:** Wristbands for X children are prepared and distributed in < 25 minutes.
 - **R1.1:** The process must support batch steps, which include preparing multiple wristbands at once rather than fully sequential actions per child.
- **O1.2:** Late-arriving children can have their wristband handled in less than 60 seconds.
- **R1.2:** Work

Goal 2: Reduce wristband-related workload using time spent as the primary measurable proxy.

- **O2.1:** Decrease overall time related to wristband-related operations.
 - **R2.1:** Total staff time spent on wristband-related tasks is lower than the current time
- **O2.2:** No staff members need to focus exclusively on wristbands (hands-on work) during field trip preparation.
 - **R2.2:** The process must be interruptible, allowing staff to switch easily between simply supervising children without losing any progress

Goal 3: The wristband application process should not pose any harm and should be easier and safer to perform in an active day care environment.

- **O3.1:** Wristband application does not require the use of any sharp tools near children
 - **R3.1:** The design must eliminate or safely isolate scissors or cutting mechanisms during application
- **O3.2:** Wristbands can be applied while children are standing or loosely queued.
 - **R3.2:** The design must function reliably without requiring children to remain seated or still for extended periods.

abstract

The Balmy Beach Community day care is a collaborative community of staff, children, and parents committed to providing quality early childhood education. Balmy Beach day care is a non-profit, school-based child care center serving children aged 4 10 12 across seven classrooms and operating within shared school spaces. Educators are the key primary stakeholders, managing split shifts, supervising rolling arrivals and departures, while maintaining structured activities and outdoor play.

Part of providing a quality education is bringing children on field trips, a source of logistical nightmares. Part of the onboarding process for these field trips is adding an identification bracelet to every child attending. This process is extremely time-consuming and puts more pressure on already strained staff members.

This RFP identifies an opportunity to automate or redesign the bracelet application process, reducing the time taken as well as the cognitive load on staff. The intent is not to replace educators, but to reduce manual handling, time pressure, and cognitive load, allowing staff to focus more fully on supervising children and maintaining safety.

Notes 2



7:30 — Centre opens (staff + first children start arriving)

Reality: arrivals are rolling and unpredictable.

- Children can arrive **whenever**, not all at once (some mornings quiet early, then a big rush later).
United Open Karate 2
- Staff begin:
 - **Signing children in on attendance**
 - Checking whether any **permission forms** / **waivers** are missing and getting them signed if needed
United Open Karate 2

Room logistics while kids arrive

- Kids are **not all placed in one room** automatically.
- Rooms open based on **capacity/maximums**: if a room is nearing max, staff **open another room** rather than cramming children into one space.

7:30–8:30 — “Build-up” period (prep begins while supervising)

Core tension: staff are trying to do admin work *and* supervise kids at the same time.
Staff must keep kids safe/engaged while parallel tasks start:

- Confirm each child has:
 - lunch / essentials
 - emergency materials ready to go
- Start gathering trip-critical items:
 - **Emergency cards** for every child going
 - **Medication** for any child who needs it on the trip
 - **Bus confirmation** copy

- Attendance list that will be **photocopied** (one copy stays at centre, one travels with the group)

8:30 — Parents are told this is the drop-off deadline

This is the *target* time families are told to arrive by for trips (Christine describes aiming for a “deadline” with a small grace window).

United Open Karate 4

- Staff try to build structure: “be in by X time,” because the group is managing a very large number of children and can’t safely absorb last-minute arrivals.

United Open Karate 4

- Christine also notes that for safety, **parents can’t drop children at the trip location** (because the child might not be in a group, emergency card might not be brought, etc.).

United Open Karate 4

8:45 — Planned bus departure

This is the “ideal” departure time tied to the 8:30 drop-off target (8:30 → 8:45).

United Open Karate 4

But reality: they often don’t leave until ~9:30

Why? Because **families arrive right at the deadline or slightly after**, and each late arrival triggers a chain of required tasks that cannot be skipped. Christine explains this exact dynamic (“people come at 8:31... then staff are filling in what they’re wearing after we boarded the bus”).

United Open Karate 4

This creates the big mismatch you want to emphasize:

- **Planned:** 8:30 drop-off → 8:45 leave
- **Actual:** preparation stretches → **departure closer to ~9:30**

Staffing + role assignment (who does what)

Typical field trip staffing numbers

- Patti: roughly **8 staff total** on a big trip:
 - about **6 regular staff**

- plus **1–2 additional “trip support” staff**
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How tasks are assigned (divide-and-conquer model)

Patti describes a clear division of labour so the morning doesn't collapse:

1. **Staff member #1 — Attendance lead**
 - Signs children in / finalizes attendance list.
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2. **Staff member #2 — Trip Description Form lead**
 - Completes the required “what the child is wearing today” documentation for each child.
 - This is time-consuming when numbers are large (example: ~60 children).
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3. **Staff member #3 — Bracelets / shirts / washroom lead (KEY BOTTLENECK)**
 - **Putting on trip bracelets** (and making sure trip shirts are on)
 - Often also handling **washrooms** / readiness tasks
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4. **Remaining staff — supervision + snack + helping children**
 - Serving snack / last-minute support
 - Supervising, playing, engaging kids (because someone must always be watching them)

The bracelet step (why it becomes the bottleneck you want to spotlight)

Bracelet application happens during the most chaotic window

- Bracelets must be applied **while children are arriving**, while other staff are doing paperwork, and while supervision still must be maintained.
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- Patti explicitly includes “**putting bracelets on**” as one of the major preparation tasks delegated to a staff member.
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Why bracelets create friction (in practice)

Even though it's “one small task,” it becomes a choke point because:

- It's assigned to **one person**, and it's **one-by-one** (child-by-child)
- Kids may be distracted, moving around, needing washroom, or arriving late (which forces repeated interruptions)

- The bracelet staff member gets pulled between bracelets, washrooms, and managing readiness, right at the peak rush
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Net effect: bracelets become a hands-on, serial task that slows the whole system — especially when the “final wave” of kids arrive around/after 8:30.

Final “go/no-go” checklist right before leaving

Before boarding / departure, staff must ensure:

- Attendance is completed and **photocopied** (one stays at centre; one travels)
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- **Emergency cards** for every child are packed
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- **Medication** is packed (for any child who needs it)
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- **Bus confirmation** is with the group
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- Children have lunches / essentials
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- Bracelets/shirts/readiness tasks are completed (or still being completed as boarding begins)

The morning before a field trip requires extensive preparation and tasks. There are 6-8 staff members that attend field trips. The morning goes as follows:

1. Before Field Trip / Early Morning Preparation

- Staff prepare as much as possible ahead of time.
- Packing materials, organizing forms, planning roles.
- Trip description forms are partially pre-filled (child physical descriptors). Only clothing must be recorded day-of.
- This preparation helps, but it cannot eliminate morning workload because trip description forms, bracelet application, and attendance must be done live.

2. 7:30 am - 8:30 am: Arrival Processing

- Children begin arriving on a rolling basis.
- 1 staff member is assigned to take attendance for each child.

- Since children do not arrive all at once, unpredictable arrivals make it impossible to process all children quickly.
- In this window, staff must confirm permission forms and check lunches and belongings.
- Missing forms must get completed before departure.
- Since each child must get processed individually, this further slows preparation.
- Once the room that children are gathered in reaches 30 children, a new room is opened. This creates movement between multiple rooms and supervision across the rooms adds complexity.
- 1 staff member must prepare emergency cards, medications, busy confirmation, and make an attendance copy (one stays, one travels). This mandatory safety work must be completed before leaving.
- 1 staff member is assigned to complete trip description forms. The form is pre-populated with physical descriptors, but staff must write what each child is wearing the day of. If there isn't enough time, staff sometimes take the form onto the bus and finish it during travel.
- 1 staff member is assigned to apply wristbands to each child. The bracelet includes the day care's name and phone number, as a safety precaution in case a child gets lost. This is a hands-on task that must be done child-by-child while managing movement, distractions, and late arrivals. This staff member is also responsible for washroom visits and ensuring that each child is wearing their trip shirt.
- Remaining staff supervise children, provide snacks, and manage behaviour. This limits the number of staff available for preparation tasks.

3. 8:30 am: Official Drop-Off Deadline

- Families are told to arrive by 8:30 am.
- However, some families arrive right at or slightly after the deadline due to various constraints within their personal lives.
- Late arrivals restart preparation steps and extend the timeline.

4. 8:45 am - Planned Departure Time

- All staff and children are intended to board the bus at 8:45 am.
- This assumes that preparation is completed within 15 minutes after the drop-off deadline.
- In practice, this is rarely achievable due to ongoing arrivals and preparation tasks.

5. 8:45 am - 9:30 am - Continued Preparation

- Due to late arrivals, staff may still be applying bracelets, documenting clothing, or dealing with last-minute issues.
- Staff are still processing children after the scheduled departure time.

- Once all children have arrived, staff confirm attendance copies, emergency cards, medications, and child readiness before leaving. As safety checks cannot be skipped, departure waits until this is complete.

6. 9:30 am: A Typical Realistic Departure Time

- All staff and children board the bus and leave.
- Departure is often significantly later than scheduled as a result of cumulative delays: rolling arrivals, paperwork, and bracelet bottleneck.

Final readiness check	Confirm attendance copies, emergency cards, medications, bus confirmation, child readiness.	Ad min + supervision staff	Must verify all documentation and safety materials before leaving.	Safety checks cannot be skipped — departure waits until complete.
~9:30 — Actual departure (typical)	Children board bus and leave.	All staff	Departure often significantly later than scheduled.	Result of cumulative delays: rolling arrivals + paperwork + bracelet bottleneck.
8:45 — 9:30 — Continued preparation	Finish bracelets, forms, attendance, supervision, last-minute issues.	All roles continue	Staff may still be documenting clothing, applying bracelets, or processing late arrivals. Some paperwork may be taken onto bus.	Primary delay window. System still processing children after scheduled departure.
8:45 — Planned departure time	Intended bus boarding time.	All staff	Assumes preparation completed within 15 minutes after drop-off deadline.	In practice, rarely achievable due to ongoing arrivals and prep tasks.

8:30	Families	All	Some families	Late
— Official drop-off deadline	told to arrive by this time.	staff	arrive right at or slightly after deadline. Late arrivals must still be processed fully.	arrivals restart preparation steps and extend timeline.

Ongoing supervision and engagement	Supervise children, provide snack, manage behaviour.	Remaining staff	Staff cannot all focus on prep — children must always be supervised and engaged.	Limits number of staff available for preparation tasks.
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The form is pre-populated with physical descriptors, but staff must write what each child is wearing **the day of**.

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If there isn't enough time, staff sometimes **take the form onto the bus and finish it during travel** (this happens specifically when arrivals run too close to departure).

Serial, hands-on task. Must be done child-by-child while managing movement, distractions, and late arrivals. Often combined with washroom duties.

Must prepare: emergency cards, medications, bus confirmation, attendance copy (one stays, one travels).

Time / Phase	What Happens	Staff Assigned	Key Operational Details	Impact on Schedule / Delays
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Before field trip day / early morning prep	Staff prepare as much as possible ahead of time (packing materials, organizing forms, planning roles).	All staff (preparation mindset)	Trip description form partially pre-filled (child physical descriptors). Only clothing must be recorded day-of.	Prep helps, but cannot eliminate morning workload because clothing and attendance must be done live.
7:30 — Centre opens	Children begin arriving on a rolling basis. Staff begin check-in.	Attendance lead + supervising staff	Children do not arrive all at once — arrival is unpredictable.	Unpredictable arrivals make it impossible to process all children quickly.
7:30 — 8:30 — Arrival processing	Sign children in, confirm permission forms, check lunches and belongings.	Attendance lead	Missing forms must be completed before departure.	Each child must be processed individually — slows preparation.
Room management during arrival	Children placed into rooms based on capacity. Additional rooms opened if numbers grow.	Supervising staff	Children are not gathered in one room . New rooms open when capacity is reached.	Movement between rooms and supervision across spaces adds complexity.
Parallel trip preparation begins	Gather required trip materials.	Admin prep staff	Must prepare: emergency cards, medications, bus confirmation, attendance copy (one stays, one travels).	Mandatory safety paperwork must be complete before leaving.

Trip description form completion	Record what each child is wearing that day.	One staff member dedicated	Must be done for every child individually (sometimes ~60 children). If unfinished, may continue on bus.	Time-consuming administrative step, especially with late arrivals.
Bracelet and physical readiness preparation (MAJOR STEP)	Put trip bracelets on children, help with washrooms, ensure trip shirts are on.	One designated staff member	Serial, hands-on task. Must be done child-by-child while managing movement, distractions, and late arrivals. Often combined with washroom duties.	Major physical bottleneck. One person must prepare many children during busiest arrival window.
Ongoing supervision and engagement	Supervise children, provide snack, manage behaviour.	Remaining staff	Staff cannot all focus on prep — children must always be supervised and engaged.	Limits number of staff available for preparation tasks.
8:30 — Official drop-off deadline	Families told to arrive by this time.	All staff	Some families arrive right at or slightly after deadline. Late arrivals must still be processed fully.	Late arrivals restart preparation steps and extend timeline.
8:45 — Planned departure time	Intended bus boarding time.	All staff	Assumes preparation completed within 15 minutes after drop-off deadline.	In practice, rarely achievable due to ongoing arrivals and prep tasks.
8:45 —9:30 — Continued preparation	Finish bracelets, forms, attendance, supervision, last-minute issues.	All roles continue	Staff may still be documenting clothing, applying bracelets, or processing late arrivals. Some paperwork may be taken onto bus.	Primary delay window. System still processing children after scheduled departure.

Final readiness check	Confirm attendance copies, emergency cards, medications, bus confirmation, child readiness.	Ad min + supervision staff	Must verify all documentation and safety materials before leaving.	Safety checks cannot be skipped — departure waits until complete.
~9:30 — Actual departure (typical)	Children board bus and leave.	All staff	Departure often significantly later than scheduled.	Result of cumulative delays: rolling arrivals + paperwork + bracelet bottleneck.
During trip (context)	Staff supervise continuously; breaks are minimal or unavailable.	All staff	High supervision responsibility continues off-site.	Demonstrates intensity of full field trip day workload.

Questions:

- is a constraint that we can't change the wristband itself to make it a sticker or smth?
- ask about 7.3 → safety stuff? Parallelization?

Have one objective for preparation, one objective for application.

	Description	Metric	Target	Justification
Goal 1	The design should enable efficient wristband preparation and application during the morning arrival period. Justification....			

Objective 1.1 (O1.1)	Device setup and loading time at the day care centre shall be ≤ 2 minutes of active staff time per trip (for up to 90 wristbands)	Minutes (mins)	< 2	Currently, staff can begin applying wristbands immediately from a pile (no dedicated pre-prep step). Any device that requires meaningful pre-loading would increase workload and be unlikely to fit into a morning where staff are already splitting attention between attendance, trip paperwork, washrooms, and supervision.
Objective 1.2 (O1.2)	The design shall allow wristband application < 10 seconds.	Seconds (s)	< 10	-one current and one former day care member stated that it can take up to 1 minute for a staff member to apply the wristband if the child is being difficult. -studies show that the average wristband application takes 15 seconds, so the design must be better than that The wristband application process was chosen to take < 34 seconds because according to Methods Time Measurement (MTM) analysis, the process of applying an adjustable snap-closure ID wristband to a person who does not resist the application takes 34 seconds. However, one current and one former day care member stated that it can take up to one minute for a staff member to apply the wristband if the child refuses to stay still. Therefore, an effective design must take less time than the current amount of time it takes for staff to apply a wristband to a cooperative child.

	Description	Metric	Target	Justification
Goal 1	The design should enable efficient wristband preparation and application during the morning arrival period. Justification....			

Objective 1.1 (O1.1)	The design shall support the preparation and distribution of wristbands for up to 90 children in < 25 minutes.	Minutes (mins)	< 25	
Objective 1.2 (O1.2)	The design shall allow wristband application < 60 seconds.	Seconds (s)	< 60	

<https://www.balmybeachdaycare.ca/>

<https://www.verazial.com/en/patient-identification-wristbands-are-they-enough-to-ensure-patient-safety/>

https://www.amazon.ca/Ouchan-Neon-Blue-Vinyl-Wristbands/dp/B06W9LMVC5/ref=asc_df_B06W9LMVC5?mcid=028489e1a3a939f7bc4fb815e93e2808&tag=googleshopc0c-20&linkCode=df0&hvadid=706846770647&hvpos=&hvnetw=g&hvrnd=17785932809171770950&hvpone=&hvpstwo=&hvmmt=&hvdev=c&hvdvcmld=&hvlocint=&hvlocphy=9197976&hvtargid=pla-895144649675&hvocijid=17785932809171770950-B06W9LMVC5-&hvexpln=0&gad_source=1&th=1

https://kaizen.com/insights/smed-reduce-changeover-boost-efficiency/?utm_source=chatgpt.com

<https://www.sugoyaindia.com/methods-time-measurement/>

The goal is to create a solution that helps educators complete necessary tasks more efficiently and safely.

[12]

MISSION STATEMENT

Balmy Beach Community Day Care Centre’s mission is to provide a safe, positive, encouraging, stimulating and bias-free learning and care environment. Our goal is to fulfill the needs of the individual as well as the group.

[13]

In hospital settings, one of the core principles is to ensure **patient safety** and avoid harm. However, basic errors—like **incorrect identification**—still occur daily, even in advanced healthcare systems, leading to serious clinical, ethical, and economic consequences.

Among the most widely used patient identification methods are **wristbands or ID bracelets**. Their use responds to the need for a visible, practical, and accessible tool to ensure that each patient receives the correct treatment. These bands typically include information such as the patient’s full name, date of birth, medical record number, and in some cases, a barcode or chip for digital reading.

[14]

Office Products > Office & School Supplies > Labels, Indexes & Stamps > Identification Badges & Accessories > ID Wristbands



VIDEO



Ouchan Neon Blue Vinyl Wristbands - 100 Pack Plastic Wristbands for Party Events

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[15]

Optimize with OTED for single-touch changeovers

OTED (One-Touch Exchange of Die) is an advanced extension of the SMED methodology, aiming to achieve setups that require only one movement or a simple gesture, usually completed in under one minute. This concept takes the quick-change principle to its most extreme level, focusing on automation, total standardization, and technical integration.

Key features of an OTED setup include:

- Changeover is performed with a single touch—via a lever, button, or command.

[16]



In today's competitive landscape, both manufacturing and service industries are under immense

pressure to deliver higher productivity, lower costs, and faster turnaround times. With global supply chains becoming more complex and customer expectations rising, businesses cannot afford inefficiencies in their processes. This is why Methods Time Measurement has become an indispensable tool for organizations that want to gain an edge in operational excellence.

MTM is not just another productivity tool—it is a scientifically developed predetermined motion time system (PMTS) that provides a universal method for analyzing work. Unlike traditional time-and-motion studies that depend heavily on a stopwatch and subjective observation, MTM eliminates guesswork by relying on a catalog of basic human motions, each with a standard time value. These building blocks allow businesses to accurately calculate how long any task should take, regardless of who performs it.

Conducting an MTM time study offers multiple advantages. It enables engineers and managers to establish precise time standards for tasks, design optimal workflows, and uncover inefficiencies that often go unnoticed. For instance, in an assembly line, MTM can highlight unnecessary hand movements, awkward reaches, or poorly placed tools—issues that may appear minor individually but collectively lead to significant productivity losses. By addressing these inefficiencies, companies can achieve better workplace design, consistent output, and reduced operational costs.

[17]

How does testing work?

Performing a test for edge sharpness is simple. Run the tester at least 50 mm (2 inches) along the edge of the subject item and then back to its starting position without the removal of the tester from the edge. Per UL 1439, the tester and corresponding tape caps are calibrated with a force to simulate human touch. As a result, a dangerously sharp edge will cut through the two outer layers of the sensing tape, visibly exposing the lowest layer of tape (black) cap. If this base layer is not exposed then the test subject's edge is acceptable for consumer use.

What kinds of materials can be tested?

Just about any accessible edge surface can be tested. The testing procedure is designed to exert a specific amount of pressure on the item in order to provide consistent and accurate results. As a result, items with metal, glass and hard plastic edges make the best test subjects.

[18]

3 Worldwide and regional design ranges

Design ranges presented in this part of ISO 7250 are directly related to ISO/TR 7250-2.

Design ranges are specified as smallest to largest 1st to 99th and 5th to 95th percentile value observed in national populations. Each percentile value in a design range comes from a single member body. Typically, product design requirements specify a smallest to largest range of values to be accommodated. However, some product standards specify medium size or midrange values, so midrange estimates are also provided here.

The worldwide ranges specified in this part of ISO 7250 are normative for product standards. However, some users of this part of 7250 may be developing products designed for a more focused, regional market. For this reason, a method is presented for creating regional design ranges from individual member body data presented in ISO/TR 7250-2. Examples of this process are shown in informative [Annex A](#). For the anthropometric data used in deriving regional and worldwide ranges, the procedure and information for acquiring these data directly follow from the process described in ISO/TR 7250-2. The process of specifying design ranges defined in this part of ISO 7250 starts from the individual member body data presented in ISO/TR 7250-2.

Similarly, for many applications, user accommodation is maximized by considering male and female design dimensions separately. For example, maximum accommodation would be achieved by designing to a range that extends from the smallest 5th percentile value, male or female, to the largest 95th percentile value, male or female. [Annex B](#) shows the results of this procedure and should be used when the largest accommodation range is desired.